1. Ethics and Ownership

- Ethics: a system of moral principles that guide behaviour based on philosophical views
- Computer Ethics
  - Regulate how computing professionals should make decisions regarding professional & social conduct.
  - A computing professional can be ethically guided by joining a professional, ethical body such as the BCS and IEEE, which have codes of conduct.
  - The BCS Code of Conduct provides the following guidance:
    - **Public Interest:** Both codes emphasize the importance of acting in the public interest and safeguarding the health, safety, and welfare of the public.
    - **Judgement and Integrity**: Professionals are expected to exercise sound judgement and maintain their integrity in their work.
    - **Ethical Practice**: Both codes stress the importance of ethical behaviour and decision-making.
    - **Lifelong Learning**: The IEEE code explicitly mentions the need for continuous learning, which aligns with the need to maintain professional competence.
  - The IEEE Software Engineering Code of Ethics defines eight principles:
    - **PUBLIC**: Software engineers shall act consistently with the public interest.
    - **CLIENT AND EMPLOYER**: Software engineers shall act in the best interests of their client and employer, while also considering the public interest.
    - **PRODUCT**: Software engineers shall ensure that their products meet the highest professional standards possible.
    - **JUDGEMENT**: Software engineers shall maintain integrity and independence in their professional judgement.
    - **MANAGEMENT**: Software engineering managers and leaders shall promote an ethical approach to the management of software development and maintenance.
    - **PROFESSION**: Software engineers shall advance the integrity and reputation of the profession, consistent with the public interest.
    - **COLLEAGUES**: Software engineers shall be fair to and supportive of their colleagues.
    - **SELF**: Software engineers shall participate in lifelong learning and promote an ethical approach to their practice.
- Ownership
  - Data ownership: having legal rights and complete control over a single piece or set of data elements.
  - Copyright gives the creators of some types of media rights to control how they're used and distributed.
  - Competitors can steal programming ideas and methods; software can easily be copied and bootlegged (sold illegally); hence, legislation is needed to protect the ownership, usage, and copyright of data.

Software Licencing

- Free Software Foundation:
  - A license gives users freedom to run, copy, distribute, study, change, and improve the software.
  - Condition: any redistributed version of software must be distributed with original terms of free use, modification, and distribution (aka copyleft)
- The Open Source Initiative:
  - The source code of open-source software is readily available to users under copyright; it does enable users to re-distribute the software.
  - The concept of an open-source program relies on the fact that users can review source code to eliminate bugs in it
- Shareware:
  - Demonstration software that is distributed for free but for a specific evaluation period only
  - Distributed on a trial basis and with an understanding that sometime later, a user may be interested in paying for it
  - Used for marketing purposes
- Commercial: Requires payment before it can be used, but includes all program's features, with no restrictions

Artifical intelligence (AI)

- Artificial Intelligence (AI): the ability of a computer to perform tasks in such a way that is conventionally associated with human intelligence:
  - AI can learn from past mistakes
    - they adapt to stop the same problem from occurring again
    - they learn to predict what might happen & raise alert
  - AI can learn to work more efficiently
    - when an action slows the system down, it can prevent this from happening again
    - when an action increases the speed of the system, it can repeat this when necessary to improve efficiency
- AI Applications
  - Developing autonomous mechanical products
  - Machine learning through data sets

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- AI Impacts
  - Social
    - Replacement of manual labour with automation could lead to massive unemployment.
    - However, it could lead to increased leisure time.
  - Economic: Due to increased innovation and efficiency with automation provided by AI, there'd be lower manufacturing costs in general
  - Environmental: Detrimental impact on the environment due to robot manufacture with limited resources and its waste disposal

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